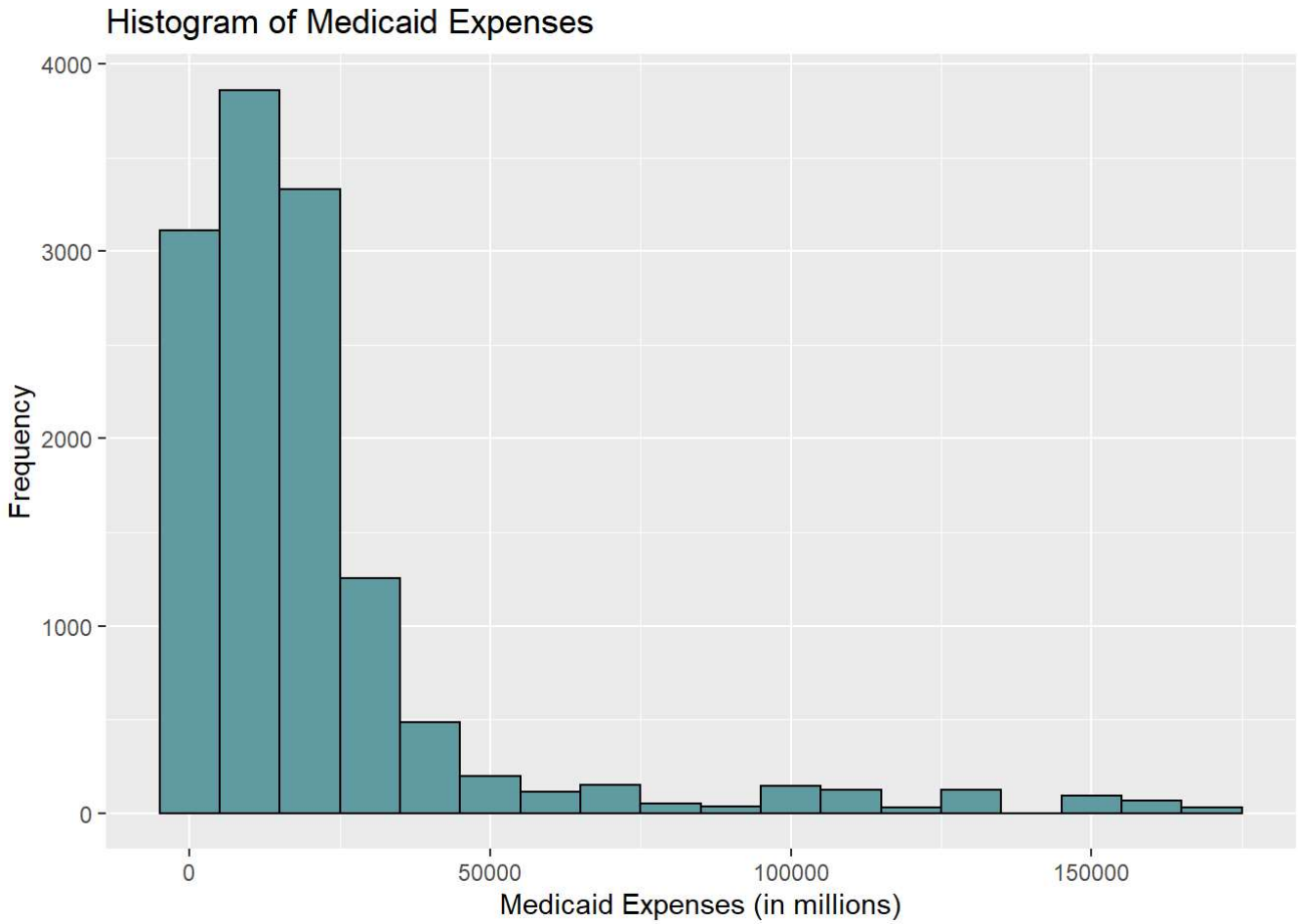
Chronic_Diabetes_Analysis

2025-07-03

Medicaid Expenses Distribution

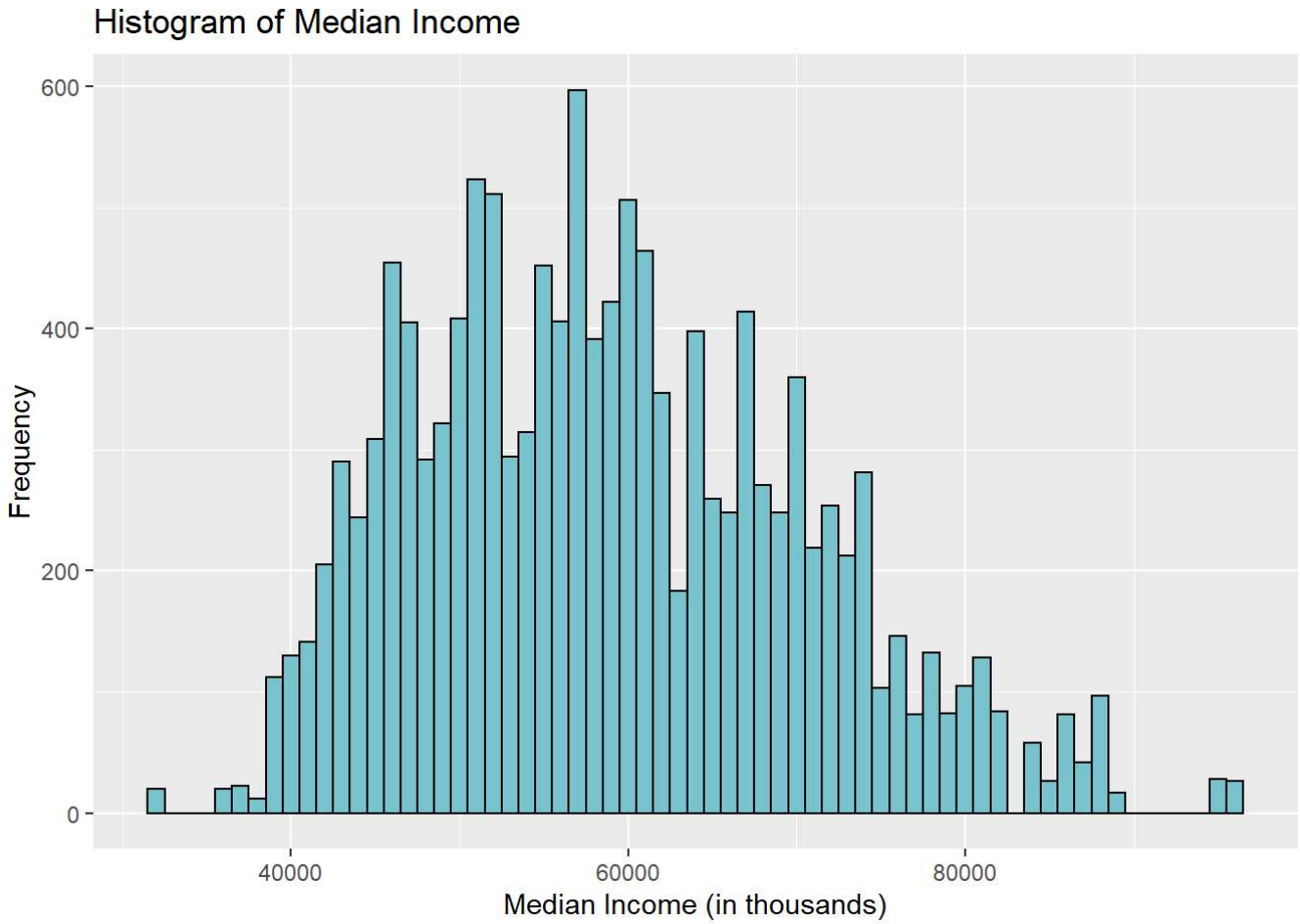
```
ggplot(final_df, aes(x = Medicaid_Expenses)) +
  geom_histogram(binwidth = 10000, fill = "cadetblue", color = "black") +
  labs(title = "Histogram of Medicaid Expenses", x = "Medicaid Expenses (in millions)", y = "Fre
quency")
```



###Insights### 1. Right-Skewed Distribution (Positive Skew): Most Medicaid expenses are concentrated at the lower end, particularly below $50,000 million. There are fewer but significant outliers with very high expenses (up to ~170,000 million), causing a long tail on the right. 2. Majority of Observations are Low to Moderate: A large portion of the data falls between 0 and 30,000 million, suggesting that many regions (or time periods) have relatively modest Medicaid spending. 3. Potential Outliers / High-Cost Areas: The sparsely populated bars beyond 100,000 million likely represent states or timeframes with unusually high costs, possibly due to: - Larger populations - Expanded Medicaid programs - High-cost procedures or demographics

Median Income Distribution

```
ggplot(final_df, aes(x = Median_Income)) +
  geom_histogram(binwidth = 1000, fill = "cadetblue3", color = "black") +
  labs(title = "Histogram of Median Income", x = "Median Income (in thousands)", y = "Frequenc
y")
```



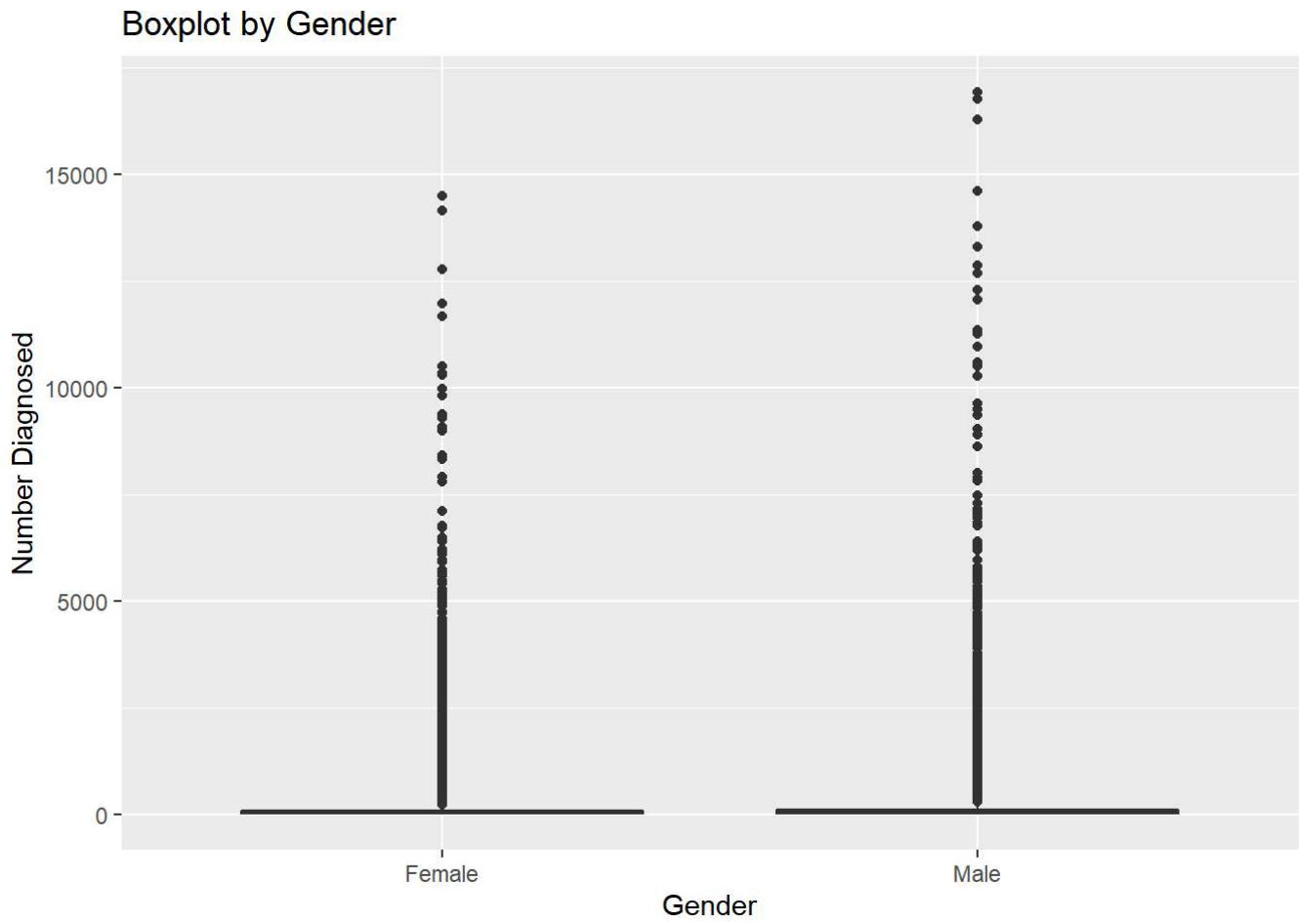
###Insights###

1. Approximately Normal Distribution with Slight Right Skew: The distribution appears roughly bell-shaped, peaking between $50,000 and $60,000. There's a slight skew to the right, indicating some regions have notably higher median incomes (above $80,000), but they are less common.
2. Most Common Range: The majority of observations fall between $45,000 and $70,000, indicating that most regions have moderate median incomes.
3. Fewer Regions with Very Low or Very High Incomes: Few regions fall below $40,000 or above $85,000, showing that extremely low or high-income areas are rare.

Diagnosed Cases by Gender

```
# filter to gender
gender <- final_df %>% filter(StratificationCategory1 == "Gender")

# number diagnosed by gender
ggplot(gender, aes(x = Stratification1, y = Number_Diagnosed)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Boxplot by Gender", x = "Gender", y = "Number Diagnosed")
```

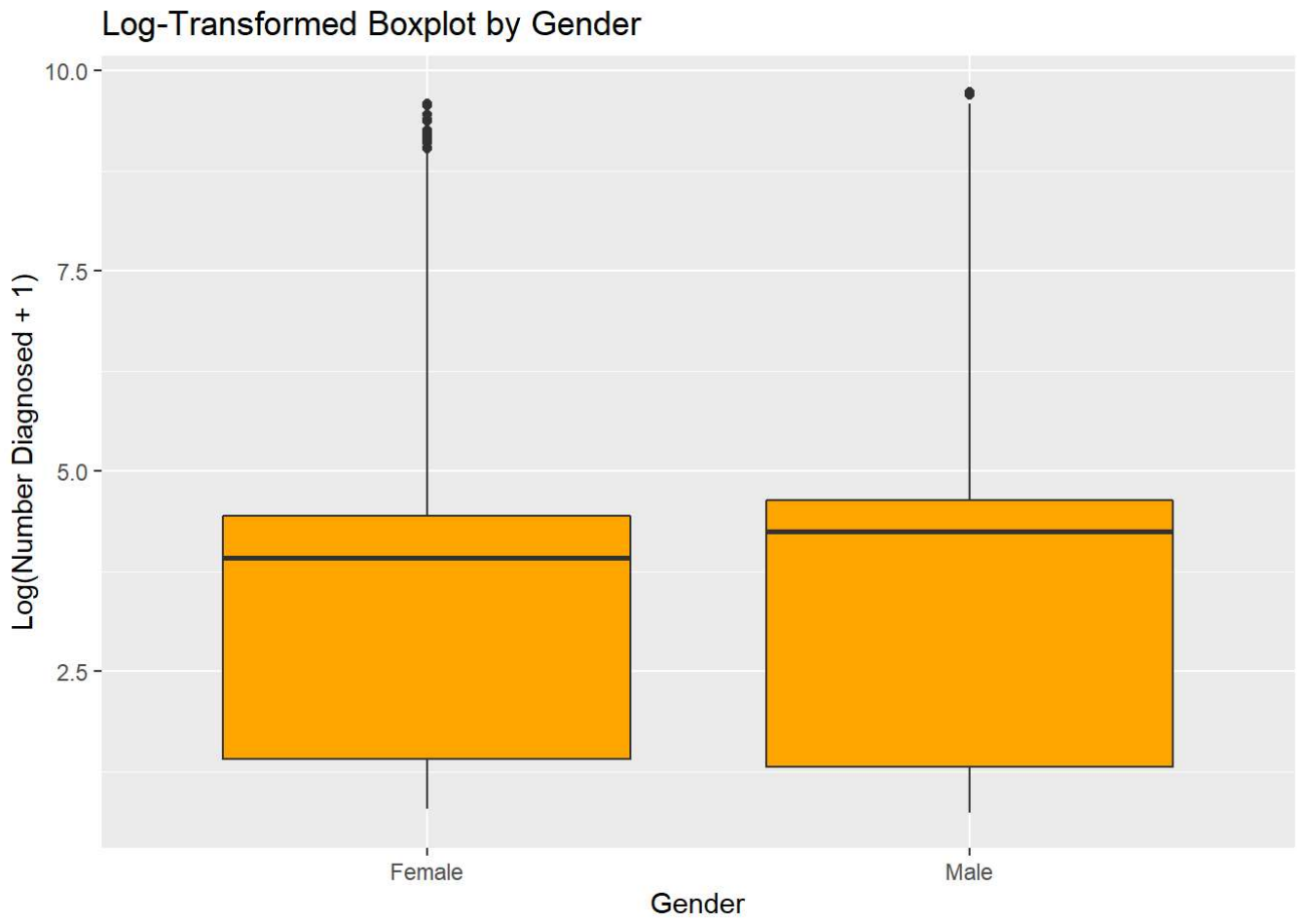


###Insights###

1. Distributions Are Similar Across Genders: The overall shape and spread of the number diagnosed appear very similar for both females and males. This suggests that the distribution of diagnoses is not heavily gender-skewed.

Diagnosed Cases by Gender with log transformation

```
ggplot(gender, aes(x = Stratification1, y = log1p(Number_Diagnosed))) +
  geom_boxplot(fill = "orange") +
  labs(title = "Log-Transformed Boxplot by Gender", x = "Gender", y = "Log(Number Diagnosed +
1)")
```



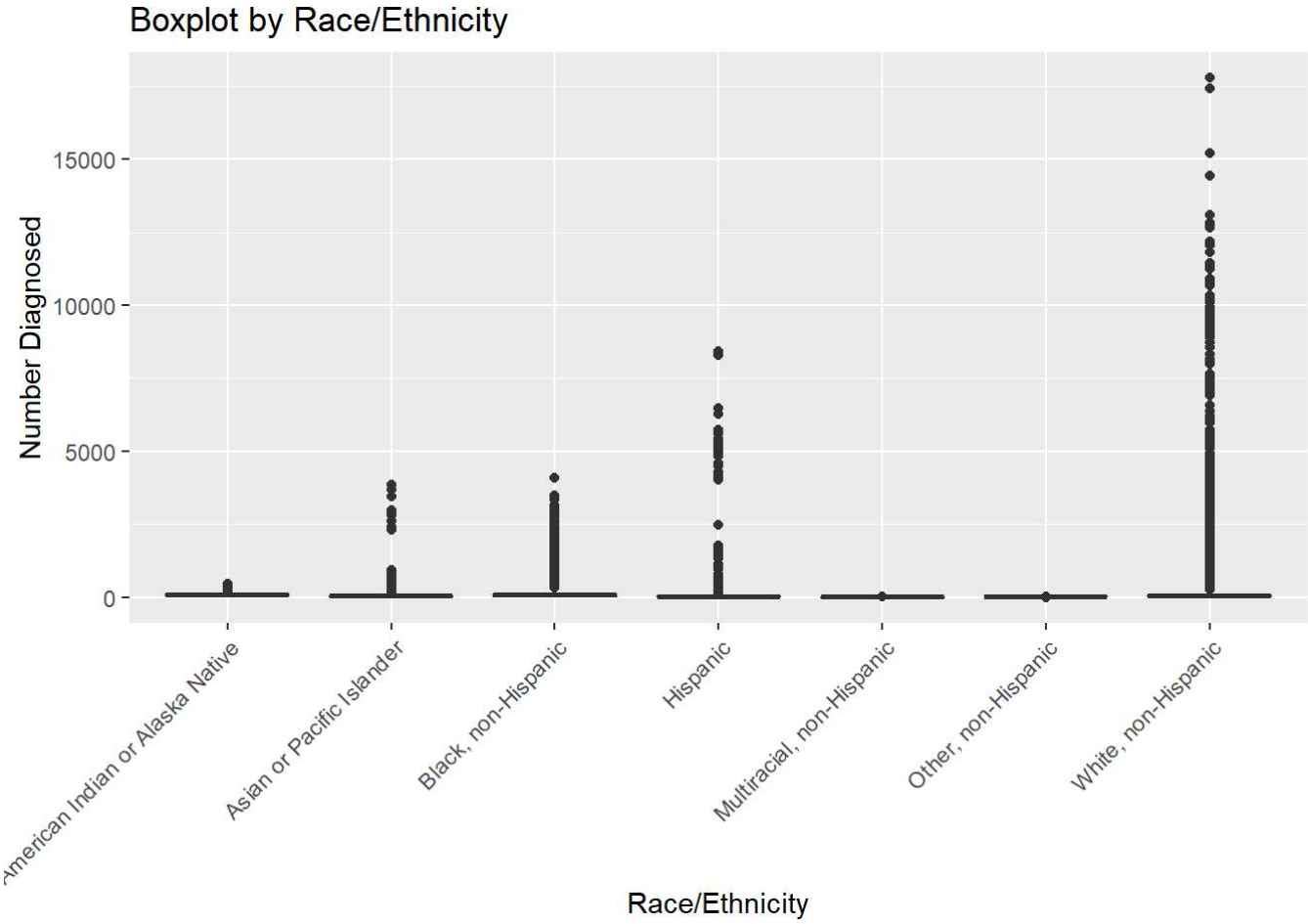
Insights

1. Comparable Distributions: Both Female and Male groups show very similar distributions after log transformation. The median values (central horizontal lines in the boxes) and interquartile ranges (IQRs) are nearly identical.
2. Presence of Outliers: Even after log transformation, both genders still show outliers above the upper whisker, indicating that a few areas still have relatively high diagnosis counts. However, the scale now reveals these outliers more proportionately.
3. Skewness Reduced: The transformation reduces the right skew, making it easier to compare gender groups without distortion from extreme values.
4. No Substantial Gender Gap: There is no significant visual difference in the number diagnosed between males and females based on this visualization.

Diagnosed Cases by Race/Ethnicity

```
race <- final_df %>% filter(StratificationCategory1 == 'Race/Ethnicity')

ggplot(race, aes(x = Stratification1, y = Number_Diagnosed)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Boxplot by Race/Ethnicity", x = "Race/Ethnicity", y = "Number Diagnosed")+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

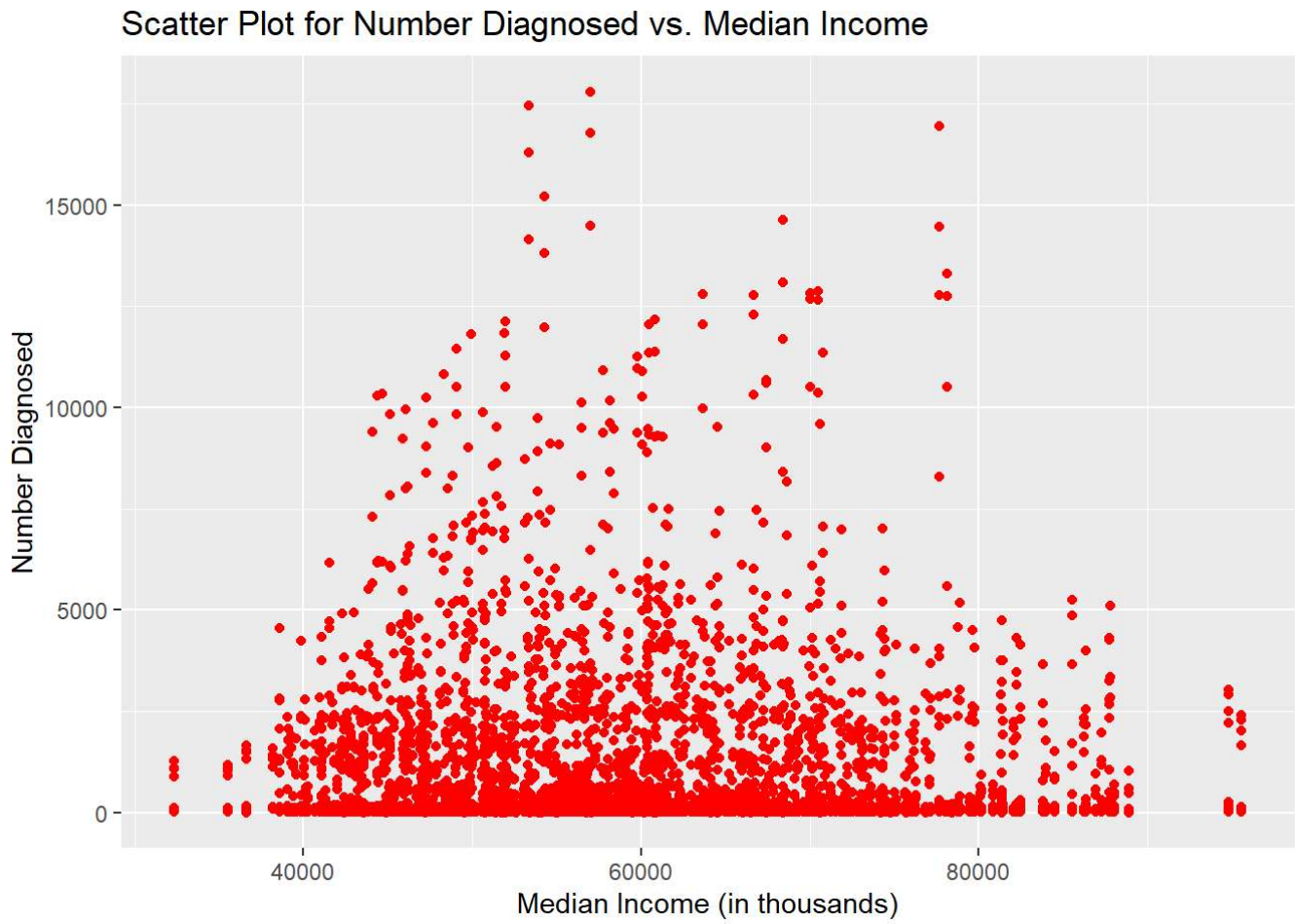


###Insights###

1. White, non-Hispanic group dominates in volume: This group has the highest number of diagnoses overall. The distribution shows a wide range, including many high-value outliers, some approaching or exceeding 400,000.
2. High variability in larger populations: Black, non-Hispanic and Hispanic groups also show a substantial range in diagnosed numbers, with numerous outliers indicating large counts in some instances. Hispanic group has some extreme outliers (e.g., near or above 200,000), suggesting high numbers in certain areas or categories.
3. Smaller groups have lower diagnosed counts: American Indian or Alaska Native, Asian or Pacific Islander, Multiracial non-Hispanic, and Other non-Hispanic groups show lower overall counts with fewer extreme values.

Diagnosed number vs median income

```
ggplot(final_df, aes(x = Median_Income, y = Number_Diagnosed)) +
  geom_point(color = "red") +
  labs(title = "Scatter Plot for Number Diagnosed vs. Median Income", x = "Median Income (in tho
usands)", y = "Number Diagnosed")
```

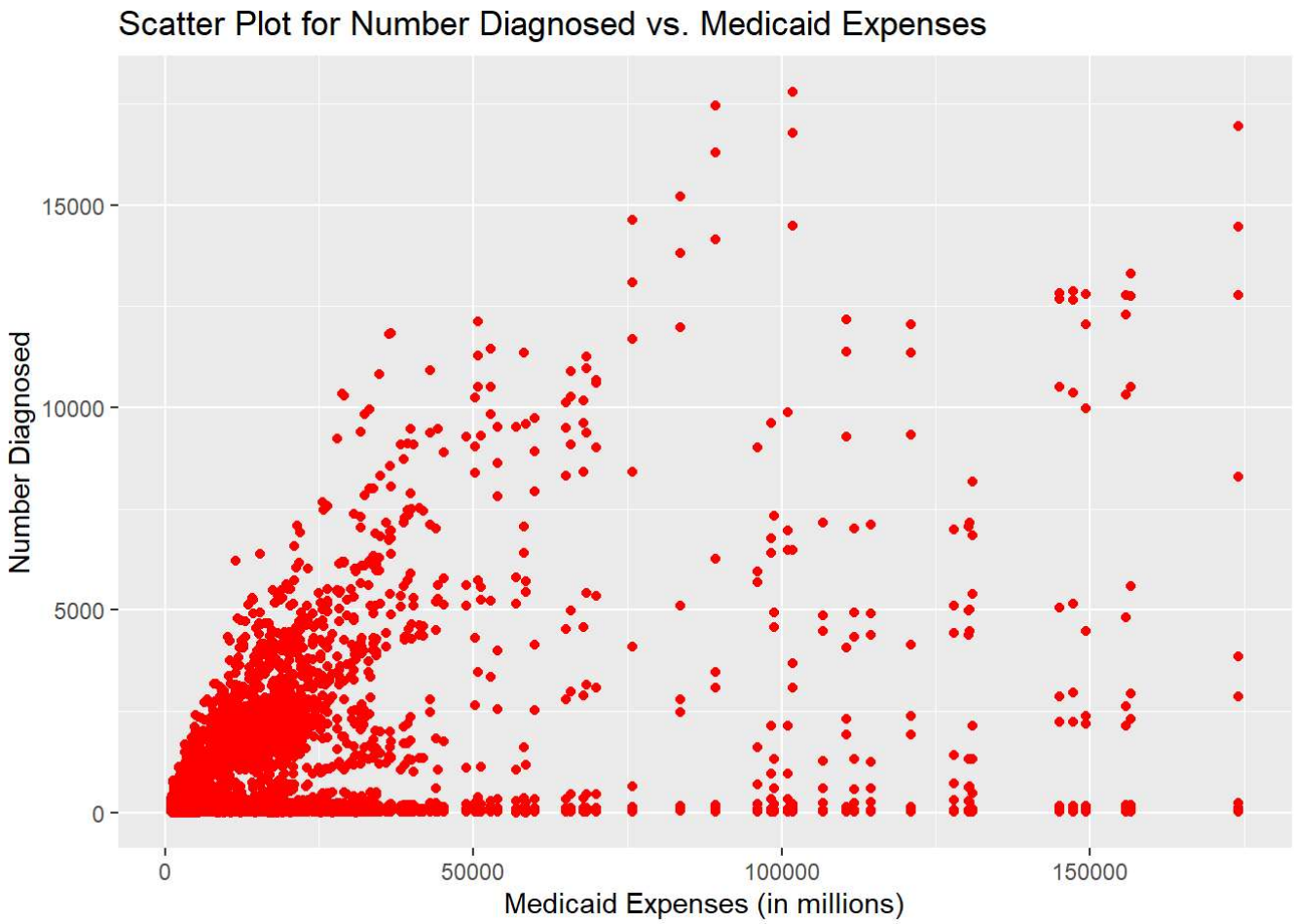


Insights

1. Inverse Relationship Observed There appears to be a negative correlation between Median Income and Number Diagnosed. Higher diagnosis counts are more common in areas with lower median incomes (under $60,000). As income increases, the number of diagnosed cases tends to decrease, though not in a perfectly linear fashion.
2. High Concentration of Diagnoses in Lower-Income Areas The densest cluster of red dots (points) lies between $35,000 and $60,000 in median income and up to 100,000 diagnoses. Several extreme outliers (200,000–400,000 diagnoses) also occur within this low-to-middle income range.
3. Fewer Diagnosed Cases in High-Income Areas For median incomes above $80,000, there are significantly fewer cases, and diagnosis counts are mostly below 50,000.

Diagnosed number vs medicaid expense

```
ggplot(final_df, aes(x = Medicaid_Expenses, y = Number_Diagnosed)) +
  geom_point(color = "red") +
  labs(title = "Scatter Plot for Number Diagnosed vs. Medicaid Expenses", x = "Medicaid Expenses
(in millions)", y = "Number Diagnosed")
```



Insights

1. Nonlinear Relationship with a Clustered Pattern There's a dense cluster of points with low Medicaid expenses (< $50,000 million) and high diagnosis counts, especially between $0 and $40,000 million. Beyond that, the relationship becomes much more scattered and unclear, suggesting other factors may influence diagnosis counts at higher expense levels.
2. Diminishing Returns Beyond a Certain Expense Level For Medicaid expenses above ~$50,000 million, the number of diagnosed cases tends to drop sharply or become inconsistent. This may indicate that beyond a certain level of Medicaid spending, other factors—like efficiency of healthcare delivery or population size—become more important than raw expense.
3. High Diagnoses at Moderate Spending Most of the very high diagnosis counts (200,000–400,000) occur within low-to-moderate Medicaid expense ranges (e.g., $10,000 to $50,000 million).

---

correlation analysis

cor_matrix <- final_df %>% select(where(is.numeric)) %>% cor(use = "complete.obs")

corrplot::corrplot(cor_matrix, method = "circle")